POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Computer Graphics (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What do you mean by Computer Graphics? Explain the working principle of refresh CRT with figure. 7
   b) What are the benefits of using a GPU for graphical rendering? What makes a GPU more ideal for graphical rendering than using a CPU? 8
2. a) Mention the limitations of DDA algorithm. Derive the necessary expression of decision parameters to rasterize a line with m<1. 8
   b) Digitize one octant of a circle by using midpoint circle generation algorithm center at origin and radius is 12. 7
3. a) A graphics designer needs to animate the swing of pendulum in center of display window with string attached at pivot point (350,300). If pendulum is displaced by 45 degrees from its resting position. Calculate the extreme points of bob's trajectory if string length is of 10 units. 8

   OR

   Reflect the triangle A(4,3), B(5,5) and C(4,6) about an axis defined by equation of line y=2-2x.

   b) Why is it required to map an object description from a window to a viewport? A triangle with vertices A(30,30) B(50,30) C(40,35) is required to be mapped from a window with its lower left corner at (20,20) and upper right corner at (60,40) to a viewport with its lower left corner at (50,50) and upper right corner at (80,70). 7
4. a) Derive composite matrix for rotation about an arbitrary axis in 3D. 8

   OR

   What are non-planar surface? How can a Bezier surface be generated from Bezier curve?

   b) Rotate the triangle with vertices A(3,3) , B(5,3) , C(4,5) by 45° in anti clockwise direction While keeping B (3,3) fixed. 7

5. a) What role does Visible Surface detection play in displaying realistic scenes? Explain Depth Sorting approach for removing hidden surfaces. 8
   b) What is Gouraud shading? Explain it with an example. What are its drawbacks? How such drawbacks are removed? 7
6. a) What is OpenGL? Write any five GLUT call back functions with their necessary syntax. 7
   b) Why is it required to visualize data sets? Explain how data sets can be visualized? 8
7. Write short notes on: **(Any two)** 2×5
   a) 2D Viewing pipeline
   b) Perspective Projection
   c) Window to viewport transformation

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